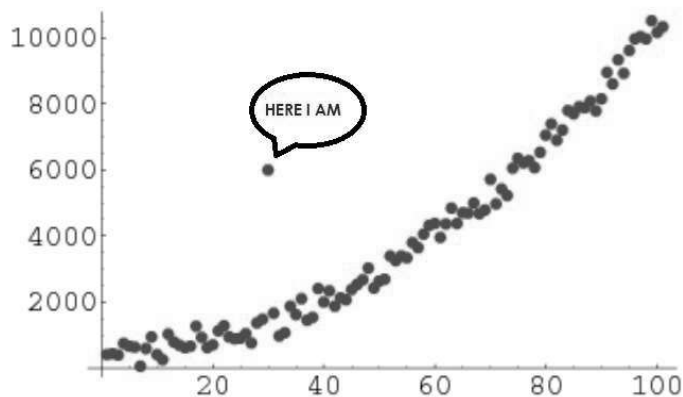


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Learnbay

OUTLIERS

What are Outliers?



“An Outlier is that observation which is significantly different from all other observations.”

What Is the Outlier Formula?

The outlier formula — also known as the 1.5 IQR rule — is a rule of thumb used for identifying outliers. Outliers are extreme values that lie far from the other values in your data set.



The outlier formula designates outliers based on an upper and lower boundary (you can think of these as cutoff points). Any value that is $1.5 \times \text{IQR}$ greater than the third quartile is designated as an outlier and any value that is $1.5 \times \text{IQR}$ less than the first quartile is also designated as an outlier.

How to identify outliers using the outlier formula:

Anything above $Q3 + 1.5 \times \text{IQR}$ is an outlier

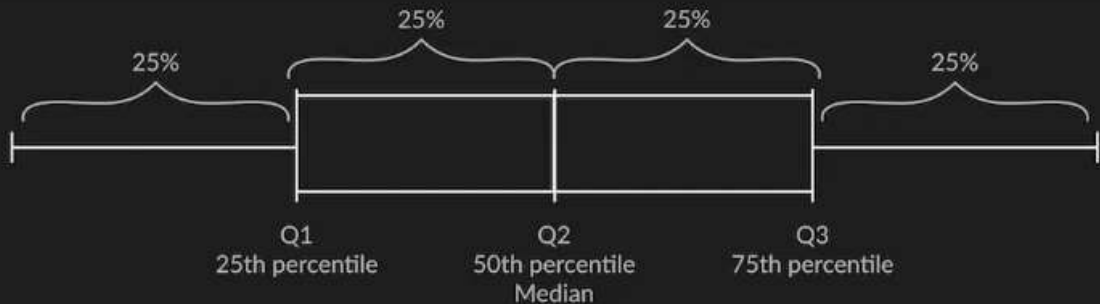
Anything below $Q1 - 1.5 \times \text{IQR}$ is an outlier

The Interquartile Range (IQR) is the distance between the first and third quartile. Subtract the first quartile from the third quartile to find the interquartile range. $\text{IQR} = Q3 - Q1$

What Are Q1, Q3, and IQR?

To use the outlier formula, you need to know what quartiles (Q1, Q2, and Q3) and the interquartile range (IQR) are.

Quartiles (Q1, Q2, Q3) divide a data set into four groups, each containing about 25% (or a quarter) of the data points. There are three quartiles: Q1, Q2, and Q3. Q1 (also known as the first quartile or lower quartile) is the 25th percentile of the data. Q2 (the second quartile) is the 50th percentile or median of the data. Q3 (the third or upper quartile) is the 75th percentile of the data.



Calculate the Outlier manually

How to Find Outliers in a Data Set

Sample Data (n=10)

27, 2, 22, 29, 19, 30, 32, 59, 52, 35

Step 1

Arrange the data in order from smallest to largest.

2, 19, 22, 27, 29, 30, 32, 35, 52, 59

Step 2

Calculate the median

Median = $(29 + 30) / 2 = 29.5$

Step 3

Find the first quartile, Q1.

To find Q1,

First Find the locator value
multiply $25/100$ by the total number of data
points (n).

$$L = n * 25/100 = 10 * 25/100 = 2.5$$

L is not a whole number, round L up to the
nearest whole number and find the
corresponding value in the data set.

That will be the first quartile.

(nearest whole no. of 2.5 is 3)

$$L = 3$$

The third value in the dataset is 22.

$$Q1 = 22.$$

(Pl. note if L is a whole no. say 2 than we will
take average of 2nd and 3rd value of the data set
as Q1 that means $(19+20)/2=19.5$,
Same concept applies for Q 3 calculation also).

Step4

Find the third quartile, Q3.

Find the locator

$$L = n * 75 / 100 = 10 * 75 / 100 = 7.5$$

Roundup nearest whole no. is 8

$$Q3 = 35.$$

$$IQR = 35 - 22 = 13.$$

Step 5

Find the upper boundary and lower boundary

$$\text{Upper boundary} = Q3 + 1.5 * IQR = 35 + 1.5 * 13 = 54.5$$

$$\text{Lower boundary} = Q1 - 1.5 * IQR = 22 - 1.5 * 13 = 2.5$$

Step6

Identify the outliers

Outliers are those data point which are more than Upper boundary and less than lower boundary.

Here 2.5 and 54.5 are boundary points.

2, 2.5, 19, 22, 27, 29, 30, 32, 35, 52, 54.5, 59

2 and 59 are outliers.

Example 1

The data below shows a high school basketball player's points per game in 10 consecutive games. Use the outlier formula and the given data to identify potential outliers.

Points Per Game

15, 24, 33, 48, 28, 21, 22, 51, 30, 31

Solution for Example 1

Outliers: 51. $Q1 = 22$, $Q3 = 33$, $IQR = 11$, lower boundary = 5.5, upper boundary = 49.5

Example 2

The data below shows the number of daily visitors to a museum. Use the given data and outlier formula to identify potential outliers.

Daily Visitors

732, 680, 815, 720, 693, 503, 740, 670

Solution for Example 2

Outliers: 503. $Q1 = 675$, $Q3 = 736$, $IQR = 61$,
lower boundary = 583.5, upper boundary = 815.

Note that there are only 8 data points ($n=8$).
When calculating $Q1$ and $Q3$, the locator value L is a whole number.

To find $Q1$, you need to take the average of the 2nd and 3rd values of the data set. To find $Q3$, you need to take the average of the 6th and 7th values.

Why Outliers are a very important and crucial aspect of Data Analysis?

While doing data preprocessing for a Data Science project, we always think about the extreme values present in the dataset, i.e, should we keep those values for our analysis or we have to just delete them from our dataset.

Difference between natural and non-natural outliers?

The non-natural outliers are those which are caused by measurement errors, wrong data collection, or wrong data entry whereas natural outliers could be the use case of fraudulent transactions in banking data, etc.

Important:- No matter how alert you are during the data collection, every Data Analyst has felt the frustration of finding

the outliers. Outliers are one of those problems which we come across almost every time while doing machine learning modeling.

Effect of outlier in performance?

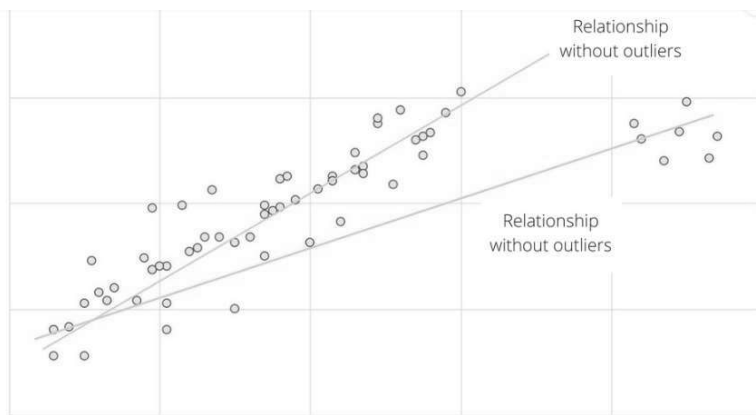


Fig. Effect of outliers on the regression line

Which statistics are affected by the outliers?

Mean → highly effected

Median → not effected

Standard Deviation → effected

Inter-Quartile Range (IQR) → not much effected by small outliers

Range → highly effected